# Portfolio Project Brief

*Driving Business Value with Data Engineering*
Advanced Software Engineering - Module 2

## How to use this brief:

**Read this document in full** and bring any questions you have on the assessment to either a workshop or the next 1:1 with your instructor. **Advance in your project journey by completing milestones**, which are progress checks on tasks that build to your overall output. Milestone submission and reviews will be part of your workshop and independent work. **Self-assess your progress** against the assessment criteria table while completing your project.

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| **Module Title** | Driving Business value with Data Engineering | | |
| **Format** | Project Evaluation Report | **Deadline** | On MyMultiverse |
| **Length** | 4000 words | **Feedback Due** | Within 20 working days of submission date |
| **Intended Audience** | | **Weighting** | 100% |
| **Project Title** | Data Engineering Project Evaluation Report | | |
| **Project Learning Outcomes** | 1. Critically evaluate the design and implementation of organisational data architecture against business initiatives. (K13, K14)<br>2. Demonstrate a fundamental understanding of how to implement a data analytics and/or engineering solution to measure and realise value. (K13, S11)<br>3. Design, implement and debug a data product. (S10)<br>4. Develop a value analysis showing the potential impact of a data-driven solution and distinctly justify your approach to stakeholders. (K13, K14) | | |
| **Project Milestones** | **Milestone 1:** Complete a detailed evaluation and restructure proposal for an existing database architecture, focusing on compliance, security, tool effectiveness, and stakeholder requirements by week 3.<br><br>**Milestone 2:** Design and develop a scalable, secure ETL pipeline proof of concept that adheres to industry best practices by week 7.<br><br>**Milestone 3:** Conduct an impact assessment of the ETL pipeline, focusing on productivity, decision-making, and ROI, complete with user feedback and visual data reports by week 9. | | |

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| **Feedback** | Written feedback and a grade will be given within 20 working days after the deadline. Opportunity for further discussion will be available during individual coaching sessions. |

## Why is this project important to me and my organisation?

This project will generate impact for your organisation in several ways.  Firstly, it significantly enhances decision-making efficiency by providing quick and informed insights through reliable and high-quality data. This leads to better strategic and operational outcomes. Additionally, optimising data processes ensures cost-efficiency, reducing both operational expenses and manual labour requirements. It also offers scalability and flexibility to handle increasing volumes and varieties of data, a vital aspect for a growing business. This project not only keeps the organisation competitive by enabling rapid data processing and analysis but also ensures adherence to regulatory standards and good governance practices.

On a personal level, it presents an opportunity for professional growth, expanding your expertise in modern technologies and strategic project management. Ultimately, this project will give you the opportunity to drive your organisation towards a more data-driven culture, future-proofing the business and fostering a collaborative, informed working environment.

## What do I need to do for this project?

You will create a **Data Engineering Project Evaluation Report** which details each step of your company evaluation, the development of a  proof of concept, and impact assessment.

First you will evaluate the current state of our organisation's data management, focusing on integration, storage, quality, GDPR/HIPAA compliance, security, and tool effectiveness. You will then identify key issues and propose actionable solutions, including a comprehensive needs analysis.You will  engage with stakeholders to understand their requirements and explore various data solutions to develop a Proof of Concept (PoC). This PoC will address specific business problems, designing an ETL process that's scalable, efficient, and compliant with security and regulatory standards. Finally, you will assess the impact of this new ETL pipeline on data processing speed, accuracy, resource efficiency, and overall business performance, making recommendations for a full-scale rollout based on user feedback and measurable improvements.

## Final Project Output

Your **Data Engineering Project Evaluation Report** will contain the following sections. Please use the descriptions in each section to guide the construction of your report, however the word count is simply a suggestion and not a strict threshold.

| Section | Content Covered |
|---|---|
| **Part 1: Current state evaluation and needs analysis** (approx. 1200 words) | • Evaluate the current state of data management at your organisation, including data source integration, storage, quality, compliance, security, and tool effectiveness.<br>• Highlight key findings and provide actionable |

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| LO1 - K13 | recommendations.<br>• Detail a comprehensive needs analysis highlighting data-related needs and pain points. |
| **Part 2: Stakeholder engagement and crafting a solution** (approx. 500 words) LO2 - S11 | • Identify key stakeholders, then gather and prioritise requirements.<br>• Using relevant data analysis techniques:<br>  ○ Analyse stakeholder feedback<br>  ○ Research and compare potential organisational data solutions.<br>• Generate ideas for developing a data solution, or update to data architecture.<br>• Use data visualization tools to present insights and gather initial stakeholder feedback. |
| **Part 3: Developing your Data Product/Proof of Concept** (approx. 1500 words) LO3 - S10 | • Understand the specific business problems to be addressed and identify all necessary data sources.<br>• Plan and define the data solution, including any ETL processes or data flows. Consider data formats, sizes, and frequency of updates in the design.<br>• Choose an appropriate data solution (see list below), focusing on compatibility and scalability. This may be a low-code, no-code, or a coded solution, depending on business needs.<br>• Develop your data product, conduct comprehensive testing, and automate with scheduling.<br>• Focus on scalability, performance optimization, security, and regulatory compliance.<br>• Ensure flexibility for future changes and adherence to best practices. |
| **Part 4: Impact assessment and recommendations for rollout** (approx. 800 words) LO4 - K14 | • Set measurable goals for your data product, aiming for improvements in data processing speed, accuracy, and resource efficiency.<br>• Monitor improvements in processing time, data quality, resource utilisation, scalability, and error rates.<br>• Assess impacts on user productivity, business decision-making, cost savings, and revenue. Collect user feedback to gauge satisfaction and ease of use.<br>• Compile findings into a comprehensive report with visualisations for clarity. |

## Examples of Projects

**Cloud-Based ETL or ELT Solutions**: Transitioning to cloud-based ETL tools can offer scalability, flexibility, and cost-efficiency. Cloud services like AWS, Azure, and Google Cloud provide robust ETL tools that can handle large volumes of data and complex transformations.

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**Data Integration and Storage Evaluation:**
Assess the current state of data integration and storage across the organization. Identify inefficiencies, data silos, and potential areas for improvement. Propose a modern data architecture that addresses these issues, such as implementing a data lake or warehouse solution that enhances data accessibility and integration.

**Stakeholder-Driven Analytics Solution:**
Engage with key stakeholders to understand their data requirements. Based on this feedback, develop a PoC for a tailored analytics solution. This could involve creating custom dashboards or implementing self-service BI tools that directly address stakeholder needs.

**Advanced Analytics Integration**: Integrating advanced analytics and machine learning capabilities into the ETL pipeline can enhance

## Project Milestones

Project milestones are completion points for the overall output of your module project. Please note that the project is intended to be a piece of applied work that develops over the course of the module, rather than a summative assessment to be completed in a short, intense period of time at the end of module learning.

**NOTE:** For step-by-step guidance on milestone completion, follow the Milestone activities for the module on your MyMultiverse account.

## Check Your Durable Skills

In addition to the assessment criteria, all assessments across your apprenticeship will also be marked on your durable skills. Use this checklist to ensure you are submitting a high quality piece of work.

- **Clear communication.** I've clearly and concisely communicated my findings and demonstrated professional language, spelling and grammar, and adhered to the word count. Please note that any referencing should be included at the end of the slide show, please refer to the style guide on Applied for further guidance.
- **Thinking strategies.** I've engaged in the critical analysis of available facts, evidence, observations, and arguments to form a judgement on what I think is important. Ultimately I've framed my analysis in a way that arrives at a solution.
- **Self-reflection.** I've considered feedback from previous work to inform the current submission. With this, I am engaging with feedback with a growth mindset in order to cultivate an attitude of continuous improvement.

## How should I submit this Portfolio Project?

Your submission should be prepared as a single PDF file. Please make sure your work adheres to the MV referencing and style guide (see MyMultiverse). All work is checked for plagiarism using Turnitin.
Ensure that you meet the MV guidelines for academic integrity by considering the following:

- **Referencing:** When discussing someone else's work, clearly acknowledge the work by using in-text citations and a reference list adhering to the Harvard referencing style.
- **Word Count:** Any text exceeding the maximum word limit will not be marked. Text that adds to the word count includes all text except the reference list or appendices. All text in diagrams and in-text citations count towards the overall word limit.
- **Submission time:** All submissions must be made by mid-day on the specified deadline.

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# How will my assessment be marked?

Carefully review the assessment criteria as you complete milestones and outputs of your project.

| Description | F | E | D | C | B | A |
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| **Critically evaluate the design and implementation of organisational data architecture against business initiatives. (K13)** | Unable to effectively evaluate the design and implementation of organisational data architecture against business initiatives. | Shows some knowledge of organisational data architecture, but is unable to apply this to a business context. | States the key elements of organisational data architecture and provides a basic evaluation of how this aligns with business initiatives. | Identifies key components of organisational data architecture and evaluates their alignment with business initiatives. Provides some recommendations for improvement. | Analyses organisational data architecture, focusing on its alignment with business initiatives. Provides clear and actionable recommendations for improvement. | Critically evaluates organisational data architecture against business initiatives. Provides a comprehensive and strategic set of recommendations for improvement, supported by clear explanations and persuasive rationale. |
| **Demonstrate a fundamental understanding of how to implement a data analytics and/or engineering solution to measure and realise value. (S11)** | Unable to identify or apply any relevant data analysis techniques to assess digital and technology solutions. | Shows limited awareness of data analysis techniques but fails to appropriately apply them to assess digital and technology solutions. | Identifies basic data analysis techniques and attempts to apply them to assess digital and technology solutions, but the application lacks depth or accuracy. | Determines and applies appropriate data analysis techniques to assess digital and technology solutions. Analyzes stakeholder feedback and compares potential solutions using basic statistical or diagnostic analysis. | Effectively determines and applies multiple appropriate data analysis techniques to assess digital and technology solutions. Conducts thorough analysis of stakeholder feedback and potential solutions using a combination of relevant techniques (e.g., statistical, diagnostic, or predictive analysis). | Demonstrates use of multiple appropriate data analysis techniques to assess digital and technology solutions. Conducts in-depth analysis of stakeholder feedback and potential solutions using advanced techniques. Effectively uses data visualization tools to present insights and gather feedback. Provides actionable recommendations based on the analysis. |

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| **Design, implement and debug a data product. (S10)** | Unable to design, implement, and debug a data product. | Shows some knowledge of the basic elements of a data product, but struggles to apply this in a meaningful way. | Able to design the basic elements of a data product, but struggles to implement or debug this in a meaningful way. | Demonstrates an understanding of the development process for a data product such as an ETL pipeline. Conducts simple tests to ensure functionality. | Develops a comprehensive data product meeting business requirements, with focus on scalability, performance, security, and compliance. Conducts thorough testing and logging. | Develops a highly adaptable, scalable data product exceeding current business needs. Implements a solution with optimized performance and proactive security measures. Provides actionable insights driving measurable business value. Evaluates the use of automation and scheduling. |
| **Develop a value analysis showing the potential impact of a data-driven solution and distinctly justify your approach to stakeholders. ( K14)** | Unable to develop a value analysis showing the potential impact of a data-driven solution and distinctly justify your approach to stakeholders. | Shows some understanding of how to develop a value analysis, but struggles to apply this to a data-driven solution or justify the approach to stakeholders. | Develops a value analysis that identifies at least one potential impact of a data-driven solution. Provides a basic justification for the approach, referencing input from at least one key stakeholder. | Develops a value analysis that quantifies 2-3 potential impacts of a data-driven solution on specific business metrics. Includes analysis of needs from at least two different stakeholder groups. Provides a justification for the approach with at least one supporting data point or case study. | Develops a value analysis that quantifies potential impact of a data-driven solution across multiple business metrics. Includes analysis of stakeholder needs from key departments. Provides a clear implementation plan with defined milestones. | Develops a multi-faceted value analysis that quantifies short-term and long-term impacts of a data-driven solution, including ROI projections. Incorporates analysis of stakeholder needs across all levels of the organization. Provides a detailed implementation plan with phased rollout strategy, specific KPIs for each phase, and contingency plans. |

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